

Compare the outcome values across the three treatment groups. Is there a statistically significant difference between the groups? Report everything I would need for a rigorous write-up.

已停止思考 >

I compared the outcome values across the three treatment groups (A, B, and C) using both a standard one-way ANOVA and a more robust Welch ANOVA because the variance assumptions were violated.

Descriptive statistics

Treatment	n	Mean	SD	Median	Min	Max
A	40	100.42	5.87	99.62	90.53	109.87
B	40	105.89	8.29	107.15	86.61	119.45
C	40	104.23	33.03	97.36	40.88	154.12

Group C showed substantially larger variability than Groups A and B.

Assumption checks

Normality (Shapiro–Wilk tests)

Group	W	p
A	0.970	0.353
B	0.929	0.015
C	0.928	0.014

The distributions for Groups B and C deviated significantly from normality.

Homogeneity of variance

Levene's test:

$F(2, 117) = 18.03, p < 0.001$

The equal-variance assumption was strongly violated.

Primary inferential result: Welch ANOVA

Because the variance assumption was violated, Welch's ANOVA is the appropriate primary test.

$F(2, 67.85) = 5.81, p = 0.0047$

Interpretation

There **was a statistically significant difference** in outcome values among the three treatment groups.

Post hoc pairwise comparisons

Welch unequal-variance t-tests were conducted with Holm correction for multiple comparisons.

Comparison	t	Adjusted p	Hedges' g	Interpretation
A vs B	-3.40	0.003	-0.75	Significant
A vs C	-0.72	0.954	-0.16	Not significant

Interpretation

- Treatment **B** produced significantly higher outcomes than Treatment **A**.
- Treatment **C** did not differ significantly from either A or B after correction.
- The A vs B effect size was moderate-to-large (Hedges' g ≈ 0.75).

Effect size

Using the classical ANOVA framework, the proportion of variance explained was:

$$\eta^2 = 0.013$$

This corresponds to a small overall effect, although the unequal variances and extreme spread in Group C complicate interpretation.

Recommended write-up

"A Welch one-way ANOVA was conducted to compare outcome values across Treatment A, Treatment B, and Treatment C. Assumptions of homogeneity of variance were violated according to Levene's test, $F(2,117) = 18.03$, $p < .001$, so Welch's ANOVA was used. The analysis revealed a significant effect of treatment on outcome values, Welch's $F(2, 67.85) = 5.81$, $p = .0047$. Post hoc Welch t-tests with Holm correction indicated that Treatment B ($M = 105.89$, $SD = 8.29$) produced significantly higher outcomes than Treatment A ($M = 100.42$, $SD = 5.87$), adjusted $p = .003$, Hedges' $g = 0.75$. No other pairwise differences were statistically significant."

